

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026
Biology — Class IX | Paper: SSC-I
Syllabus: Chapter 10 — Evolution

Total Marks: 65

Time: 2h 30m

Version: 1

General Instructions:

1. Write your Roll Number in the space provided on the answer sheet. Do NOT write your name.
2. Attempt ALL questions from Section A. In Sections B and C, attempt the required number choosing either the main question OR its alternative (OR).
3. All parts of a question must be answered in sequence and at one place.
4. Extra attempt of any question or any part thereof will NOT be considered.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	Evolution is defined as the gradual change in a population's traits over time in response to:	☐ Food supply	☐ Environmental conditions	☐ Geographic barriers	☐ Reproductive rate	
2	Which of the following is NOT listed as a source of variation in the textbook?	☐ Genetic recombination	☐ Gene flow	☐ Photosynthesis	☐ Mutation in gamete-forming cells	
3	Charles Darwin put forth the Theory of Natural Selection after a voyage on the ship:	☐ HMS Victory	☐ HMS Beagle	☐ HMS Endeavour	☐ HMS Discovery	
4	An inherited trait that increases an organism's chances of survival in specific environmental conditions is called:	☐ Variation	☐ Mutation	☐ Adaptation	☐ Speciation	
5	Natural selection is sometimes called "survival of the fittest." What most accurately measures an organism's fitness?	☐ Its mutation rate	☐ How many fertile offspring it produces	☐ Its ability to withstand extremes	☐ How much food it obtains	
6	Before 1848, almost all peppered moths (<i>Biston betularia</i>) in Britain had:	☐ Dark colour	☐ Light colour with dark patches	☐ No wings	☐ Black and white stripes	
7	The bones in the limbs of humans, dogs, birds, and whales share the same general structure. These are known as:	☐ Vestigial structures	☐ Analogous structures	☐ Homologous structures	☐ Fossil records	
8	A mutation is defined as:	☐ A change in appearance	☐ A change in behaviour	☐ A change in gene	☐ A change in environment	
9	Farmers bred Neli-Ravi buffalo to produce more milk than older breeds. The process used was:	☐ Natural selection	☐ Genetic engineering	☐ Selective breeding	☐ Adaptation	
10	Speciation is the evolutionary process by which:	☐ A new breed is formed	☐ A new species is formed	☐ A new population is formed	☐ A new race is formed	
11	Vestigial organs are structures that are:	☐ Newly evolved and active	☐ Inactive; remains from a previous common ancestor	☐ Found only in plants	☐ The same as homologous structures	
12	Natural selection operates to produce changes in:	☐ Individuals	☐ Populations	☐ Races	☐ Genes only	

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt 11 questions. Choose either the main question OR its alternative.

Part	Question	Marks	OR	Alternative Question	Marks
2(i)	Define evolution. How old do scientists believe the Earth to be, and what does this have to do with how life forms developed?	3	OR	Define variation. List any three sources of variation that arise from meiosis and related processes.	3
2(ii)	What is adaptation? How does it differ from variation? Give one example of an adaptive trait.	3	OR	Explain how variation promotes evolution within a population. What advantage do some individuals gain?	3
	State the four bases of Darwin's Theory of			What observations did Darwin make at the	

2(iii)	Natural Selection. Name each basis briefly.	3	OR	Galapagos Islands? How did the finches he found support the idea of adaptation?	3
2(iv)	When was Darwin's book "On the Origin of Species by Natural Selection" published and what theory did it describe? State the year the theory was first put forth.	3	OR	Describe what is meant by "overproduction" and "competition" in the context of natural selection theory.	3
2(v)	Describe Industrial Melanism. What was the moth population in Britain like before and after the Industrial Revolution?	3	OR	Explain why light-coloured peppered moths became easy prey after the Industrial Revolution. What role did predator birds play?	3
2(vi)	What are the three conclusions drawn from the analysis of the peppered moth case study?	3	OR	After the 1950s, England's industrial pollution was reduced. What happened to the peppered moth population as a result, and what does this show?	3
2(vii)	What is the fossil record? Name one well-documented example of evolution shown by the fossil record mentioned in the textbook.	3	OR	What are vestigial organs? Give two examples of vestigial organs from animals.	3
2(viii)	What are homologous structures? Give one example with four animals whose limb bones show homology.	3	OR	What is comparative anatomy as evidence of evolution? How does sharing the same fundamental bone shape support common ancestry?	3
2(ix)	What is artificial selection? Name the two techniques included under artificial selection.	3	OR	Give two examples of improving plants by selection. What traits do farmers select for, and how is yield affected over generations?	3
2(x)	Define speciation. What role does geographical isolation play in the formation of new species?	3	OR	What is reproductive isolation? How does it eventually lead to the existence of two distinct species from one ancestral species?	3
2(xi)	Give three differences between natural selection and artificial selection from the textbook's comparison table.	3	OR	How does the process of selective breeding improve animals? Give one example from the textbook with the desired trait that was selected.	3

SECTION C — Long Questions (4 × 5 = 20 Marks)

Part	Question	Marks	OR	Alternative Question	Marks
3(i)	Explain Darwin's Theory of Natural Selection in detail. Describe all four bases (overproduction, competition, variations, natural selection) and explain how each leads to the next step in the evolutionary process.	5	OR	Describe Darwin's observations during his voyage on the HMS Beagle. What did he observe about finch species at the Galapagos Islands, and how did this lead him to connect speciation, adaptation, and geographic isolation?	5
3(ii)	Using the peppered moth (<i>Biston betularia</i>) as a case study, explain how natural selection changed the population's colour from light to dark during the Industrial Revolution, and back toward light after the 1950s. Include the role of predator birds and tree-bark colour in each phase.	5	OR	Describe the three types of evidence for evolution: palaeontology (fossil record), comparative anatomy (homologous and vestigial structures), and selective breeding. Give one specific example for each type.	5
3(iii)	Compare natural selection and artificial selection. Use the textbook's comparison table to identify five key differences. Then give one example of each process from living organisms.	5	OR	What is artificial selection? Describe the four-step process by which artificial selection increases crop yield over generations. Then explain how animals are also improved by selection, with one example.	5
3(iv)	Explain the concept of speciation. Define speciation and describe how geographical isolation followed by reproductive isolation leads to the formation of two new species from one ancestral species. Include the three patterns of biological diversity Darwin identified.	5	OR	Describe variation and its role in evolution. Define variation, state five sources of variation, explain what adaptation is and how it relates to variation, and explain how heritable variation drives evolutionary change over generations.	5

Invigilator's Signature: _____

Candidate's Signature: _____

Chapters Covered: Chapter 10 — Evolution (Sections 10.1-10.3)

— End of Question Paper —